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DIFFERENCE-IN-DIFFERENCES WITH VARIATION IN TREATMENT TIMING

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post-treatment effect, which is closer to -5. My proposed balance test detects significantly higher per-capita income and male/female sex ratios in reform states, in contrast to joint tests of covariate balance across timing groups, which cannot reject the null of balance. Finally, my results show that much of the sensitivity across specifications comes from changes in weights, or a small number of 2x2 DD's, and need not indicate bias.

I. THE DIFFERENCE-IN-DIFFERENCES DECOMPOSITION THEOREM

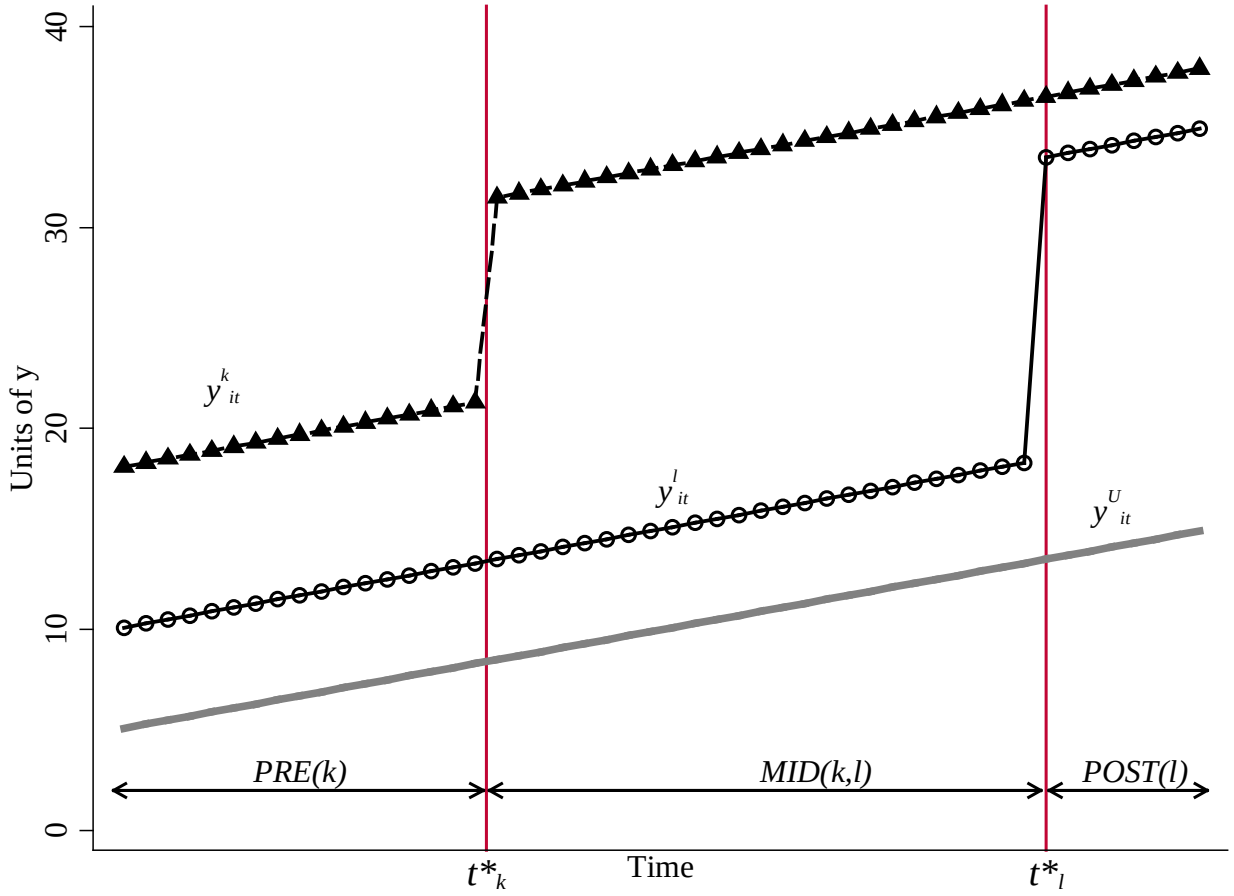
When units experience treatment at different times, one cannot estimate equation (1) because the post-period dummy is not defined for control observations. Nearly all work that exploits variation in treatment timing uses the two-way fixed effects model in equation (2) (Cameron and Trivedi 2005 pg. 738). Researchers clearly recognize that differences in *when* units received treatment contribute to identification, but have not been able to describe how these comparisons are made.⁹

The simplest way to illustrate how treatment timing works is to consider a balanced panel dataset with T periods (t) and N cross-sectional units (i) that belong to either an untreated group, U ; an early treatment group, k , which receives a binary treatment at t_k^* ; and a late treatment group, ℓ , which receives the binary treatment at $t_\ell^* > t_k^*$. Figure 1 plots this structure.

Throughout the paper I use “group” or “timing group” to refer to collections of units either treated at the same time or not treated. I refer to units that do not receive treatment as “untreated” rather than “control” units because, while they obviously act as controls, treated units do, too. k will denote an earlier treated group and ℓ will denote a later treated group. Each group's sample share is n_k and the share of time it spends treated is $\bar{D}_k$. I use $\bar{y}_b^{POST(a)}$ to denote the sample mean of y_{it} for units in group b during group a 's post period, $[t_a^*, T]$. $(\bar{y}_b^{PRE(a)})$ is defined similarly.)

⁹ Angrist and Pischke (2015), for example, lay out the canonical DD model in terms of means, but discuss regression DD with timing in general terms only, noting that there is “more than one...experiment” in this setting.

Figure 1. Difference-in-Differences with Variation in Treatment Timing: Three Groups



Notes: The figure plots outcomes in three groups: a control group, U , which is never treated; an early treatment group, E , which receives a binary treatment at $t_k^* = \frac{34}{100}T$; and a late treatment group, ℓ , which receives the binary treatment at $t_\ell^* = \frac{85}{100}T$. The x-axis notes the three sub-periods: the pre-period for group k , $[1, t_k^* - 1]$, denoted by $PRE(k)$; the middle period when group k is treated and group ℓ is not, $[t_k^*, t_\ell^* - 1]$, denoted by $MID(k, \ell)$; and the post-period for group ℓ , $[t_\ell^*, T]$, denoted by $POST(\ell)$. I set the treatment effect to 10 in group k and 15 in group ℓ .